Exercise Sheet 4

Parameter Fitting, Linear and Logistic Regression, Calibrated Scores

Probabilistic Machine Learning

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4.A A Posterior and Its Peak

Exercise 4.A.1: Two Analysts, One Campaign

Same data: 60 of 400 converted. Analyst A uses a flat prior $\mathrm{Beta}(1, 1)$; analyst B, from earlier campaigns, uses $\mathrm{Beta}(20, 80)$.

1. Write down both posteriors and their means.
2. How many "pseudo-observations" is B's prior worth? Why do the two analysts nearly agree?
3. The data arrived in two batches: 25 of the first 200, 35 of the second 200. Show that updating twice gives the same posterior as updating once.
4. What is A's probability that the next user converts?

Exercise 4.A.2: Peak, Mean and a Prior's Peak

Returning users, $\theta = P(\text{user returns})$, prior $\text{Beta}(a, b)$, h of n users return.

1. Maximise $\log p(D \mid \theta) + \log p(\theta)$ and show $\hat{\theta}_{\text{MAP}} = \frac{h+a-1}{n+a+b-2}$.
2. **0 of 4** users return. Give the MLE, the MAP under $\text{Beta}(2, 2)$ and the posterior mean under $\text{Beta}(2, 2)$. Which one would you report to a product manager, and why?
3. The posterior mean under a flat prior is $\frac{h+1}{n+2}$. Under which prior is the **MAP** equal to it?
4. After 3 of 3 with a flat prior the posterior is $\text{Beta}(4, 1)$, with density $4\theta^3$. Compute its central 95% interval.

Exercise 4.A.3: Fitting Is Counting

An attribute takes k values. Within one class we observed it n times, value v exactly n_v times ($\sum_v n_v = n$). Model it as categorical with parameters $\theta_1, \ldots, \theta_k \geq 0, \sum_v \theta_v = 1$.

1. Write down the log-likelihood $\ell(\theta)$ of the n i.i.d. observations.
2. For $k = 2$, substitute $\theta_2 = 1 - \theta_1$ and show that the MLE is $\hat{\theta}_1 = n_1/n$ — the h/n of Lecture 04.
3. For general k, maximise ℓ under the constraint $\sum_v \theta_v = 1$ (Lagrange multiplier) and show $\hat{\theta}_v = n_v/n$.
4. Why can naive Bayes fit every attribute's table separately, while a full joint $p(x \mid y)$ could not be fitted attribute by attribute?

Exercise 4.A.4: Smoothing Is a Prior

A yes/no flag was seen h times among n frauds.

1. With a $\mathrm{Beta}(1,1)$ prior on $\theta = P(\mathrm{flag} \mid \mathrm{fraud})$, what is the posterior (Lecture 04)? Show that its **mean** is Laplace's $(h+1)/(n+2)$.
2. What is the posterior's **peak** for $h = 0$? What does that say about "keep the peak"?
3. "Add-α" smoothing uses $(h+\alpha)/(n+2\alpha)$. Which prior belongs to it? Compute the estimate for $h = 0, n = 100$ and $\alpha = 0.5, 1, 10$.
4. A merchant category with $k = 10$ values; one category never occurred in 100 frauds. What does Laplace smoothing give, and which prior is behind it?

4.B Least Squares Is a Likelihood

Exercise 4.B.1: Four Flats and a Villa

Model $y = w_0 + w_1 x + \varepsilon, \varepsilon \sim \mathcal{N}(0, \sigma^2)$ i.i.d.; x in m², y in k€.

x	40	60	80	100
y	150	200	280	330

1. Write down $\ell(w_0, w_1, \sigma^2)$ and show that for **every** σ^2 the maximum over (w_0, w_1) is the least-squares fit.
2. Compute $\hat{w}_1 = S_{xy}/S_{xx}$ and $\hat{w}_0 = \bar{y} - \hat{w}_1 \bar{x}$, the residuals and the RSS.
3. Compute $\hat{\sigma}^2_{\mathrm{ML}} = \mathrm{RSS}/n$ and the unbiased $\mathrm{RSS}/(n-2)$. By which factor does the ML version underestimate σ^2 on average?
4. Give the plug-in 95% band $\hat{y} \pm 1.96\hat{\sigma}_{\mathrm{ML}}$ at 80 m² and at 600 m². Then use the half-width that counts the uncertainty about $\mathbf{w}$, $1.96\hat{\sigma}\sqrt{1 + \frac{1}{n} + \frac{(x-\bar{x})^2}{S_{xx}}}$. What do you conclude?

4.C Classification: Only p(y | x)

Exercise 4.C.1: Cross-Entropy and Its Gradient

$h_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$ with $\sigma(z) = 1/(1+e^{-z})$ and $\ell(\mathbf{w}) = \sum_i [y_i \ln h_i + (1-y_i)\ln(1-h_i)]$.

1. Show $\sigma'(z) = \sigma(z)(1-\sigma(z))$.
2. Show $\frac{\partial \ell}{\partial \mathbf{w}} = \sum_i (y_i - h_i)\mathbf{x}_i$.
3. Six transactions, $\mathbf{x} = (1, \text{new device})$: new device — fraud, fraud, legitimate; known device — fraud, legitimate, legitimate. Compute the gradient at $\mathbf{w} = 0$.
4. Set both components of the gradient to zero: which scores does the MLE give to new and to known devices? Give $\hat{w}_0$, $\hat{w}_1$ and read $\hat{w}_1$ as an odds ratio.

Exercise 4.C.2: The Peak That Runs Away (Python)

```
import numpy as np
x = np.array([-2.0, -1.5, -1.1, -0.6, -0.3, 0.2, 0.6, 1.0, 1.4, 1.9])
y = np.array([0, 0, 0, 0, 0, 0, 1, 1, 1, 1])
sigma = lambda z: 1 / (1 + np.exp(-z))

def fit(tau2=None, steps=10_000, eta=0.05):
    w0 = w1 = 0.0
    for t in range(1, steps + 1):
        h = sigma(w0 + w1 * x)
        g0 = ___                        # gradient for the intercept
        g1 = ___                        # gradient for the slope
        if tau2 is not None:
            g1 -= ___                   # log-prior N(0, tau2) on the slope
        w0, w1 = w0 + eta * g0, w1 + eta * g1
        if t in (10, 100, 1_000, 10_000):
            h = sigma(w0 + w1 * x)
            print(t, round(w1, 2), np.sum(y * np.log(h) + (1 - y) * np.log(1 - h)))
```

1. Fill in the blanks and run `fit()`. What happens to the slope, the log-likelihood and the scores? Why will it never stop?
2. Run `fit(tau2=1)` and `fit(tau2=100, steps=100_000)`. Explain the difference with the MAP formula of the lecture.

4.D Generative vs Discriminative

Exercise 4.D.1: The Same Line

Naive Bayes with yes/no flags, $\theta_{jy} = P(x_j = 1 \mid y)$, prior $\pi = P(y = 1)$. Lecture numbers: $\pi = 1/1000$; new device 0.6 vs 0.1; foreign IP 0.7 vs 0.05; night 0.5 vs 0.2 (fraud vs honest).

1. Show that $\log \frac{P(y=1|x)}{P(y=0|x)} = w_0 + \sum_j w_j x_j$ and give w_j and w_0.
2. Compute the three weights and w_0, and check that a transaction with all three flags gets $P(\text{fraud} \mid x) \approx 17.4\%$.
3. The foreign-IP flag is **missing**. Compare what naive Bayes should do with plugging $x_{\text{IP}} = 0$ into the linear form.
4. Why can logistic regression, which has the same linear form, not do what naive Bayes does in 3?

Exercise 4.D.2: Counting Twice, in Code

Complete the three marked lines (numpy only), then run the script.

```
import numpy as np
rng = np.random.default_rng(42)
n = 5000
y = (rng.random(n) < 0.3).astype(int)                        # 1 = fraud
flag = lambda pf, ph: (rng.random(n) < np.where(y == 1, pf, ph)).astype(int)
device = flag(0.6, 0.1)
copy = np.where(rng.random(n) < 0.95, device, 1 - device)   # a near-copy
night = flag(0.5, 0.2)
X = np.column_stack([device, copy, night])

def fit_nb(X, y, alpha=1.0):
    prior = ...                     # (a) P(y = 1)
    th1 = ...                       # (b) Laplace-smoothed P(x_j = 1 | y = 1)
    th0 = ...                       #     and P(x_j = 1 | y = 0)
    w = np.log(th1 / th0) - np.log((1 - th1) / (1 - th0))
    w0 = np.log(prior / (1 - prior)) + np.log((1 - th1) / (1 - th0)).sum()
    return w0, w

def fit_lr(X, y, steps=5000, lr=0.5):   # gradient ascent, as in the lecture
    Xb = np.column_stack([np.ones(len(X)), X]); w = np.zeros(Xb.shape[1])
    for _ in range(steps):
```

1. Print w_0, w and $P(\text{fraud} \mid x = (1, 1, 1))$ for both models.
2. Compare the weights of `device` and `copy`. Which model counts the device twice, and which shares the credit?
3. Which probability is closer to the fraction of frauds among the training rows with $x = (1, 1, 1)$?

4.E When Is a Score a Probability?

Exercise 4.E.1: The Rebalanced Fraud Model

A model is trained on a 50/50 sample; on real traffic fraud is **1 in 1,000**. Undersampling kept every fraud and a random share of legitimate cases.

1. Use Bayes' theorem in odds form to show: real odds = model odds × $\frac{1/999}{1}$. Which assumption about $p(\mathbf{x} \mid y)$ do you need?
2. Convert the scores 0.5, 0.9 and 0.99 into real fraud probabilities.
3. Blocking a legitimate card costs **€10**, missing a fraud **€500**. Above which real probability should a card be blocked, and which **raw score** is that?

Exercise 4.E.2: Weight Decay Is a Prior

1. With $y_i = \mathbf{w}^\top \mathbf{x}_i + \varepsilon_i, \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ and a prior $\mathbf{w} \sim \mathcal{N}(0, \tau^2 \mathbf{I})$, show that the MAP minimises $\mathrm{RSS}(\mathbf{w}) + \lambda \|\mathbf{w}\|^2$ with $\lambda = \sigma^2/\tau^2$.
2. One weight, no intercept, $y = wx + \varepsilon$. Show $\hat{w}_{\mathrm{MAP}} = \dfrac{\sum_i x_i y_i}{\sum_i x_i^2 + \lambda}$.
3. Data $(x, y) = (1, 2), (3, 7)$, $\sigma^2 = 1, \tau^2 = 0.1$. Compute the MLE and the MAP. What happens as $\tau^2 \to \infty$ and as $\tau^2 \to 0$?